Introduction to Artificial Intelligence (AI)

Introduction and history of AI

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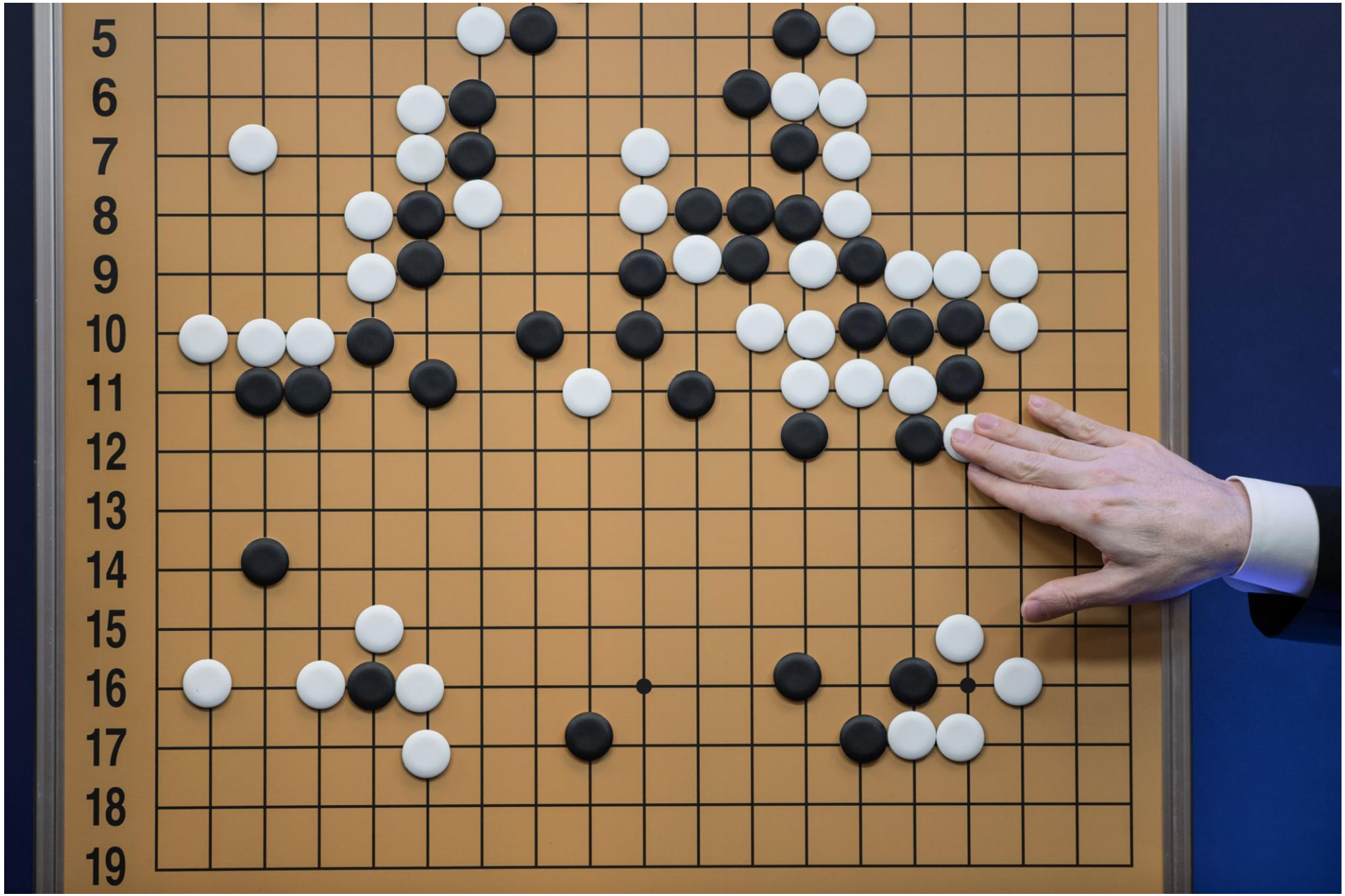
What is an AI?

Artificial Intelligence

YouTube video

Artificial Intelligence

Artificial Intelligence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AlphaGO 2016

Lee (Professional GO player) – AI (AlphaGo)
1-4

AlphaGo - The Movie | Full award-winning documentary

https://www.youtube.com/watch?v=WXuK6gekU1Y&t=1s

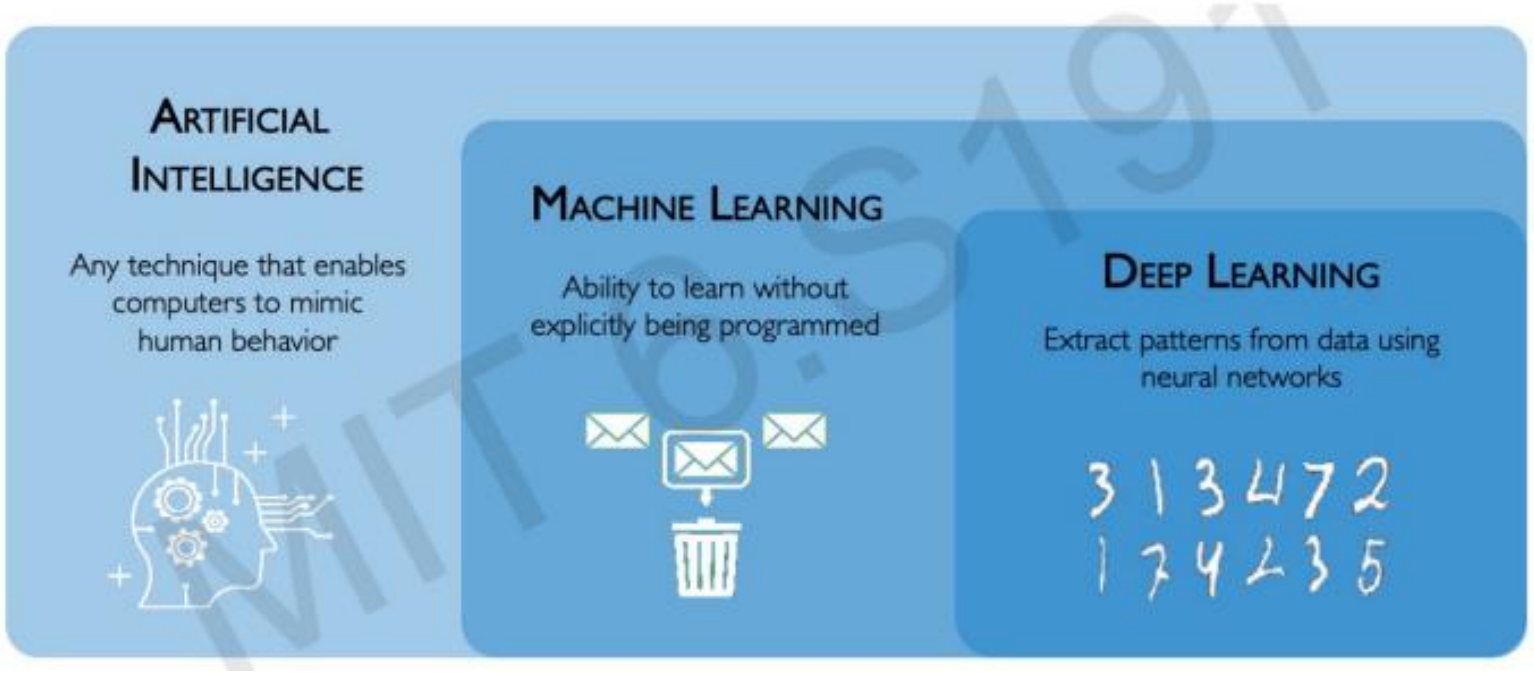
Artificial Intelligence

- Games
- Chatbots
- Recommendation systems (music, Netflix)
- Language translation
- Face recognition
- Siri, Alisa
- etc.

Artificial Intelligence

- AI is when machines show intelligence, similar to how humans think and make decisions.
- In computer science, a smart machine can understand its surroundings and make choices to help it achieve a goal.
- AI research focuses on areas like reasoning, learning, understanding language, planning, recognizing surroundings, and moving or controlling objects.

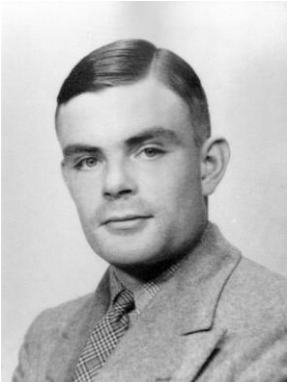
Artificial Intelligence

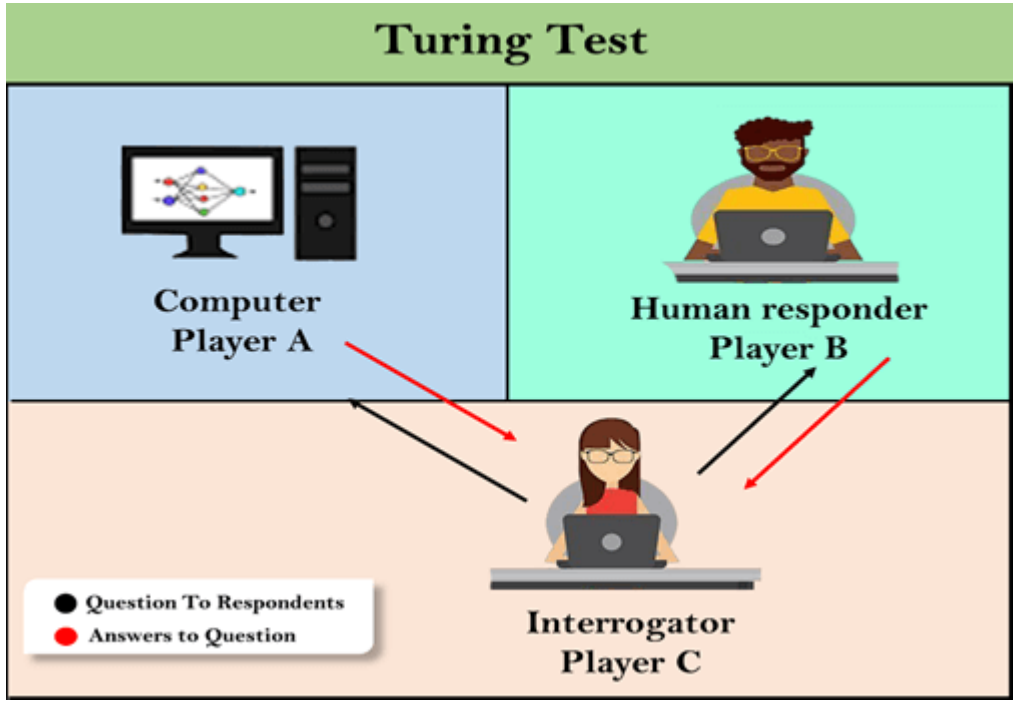


Recap

The history of AI

Alan Turing (1936): One of the first people to shape the idea of machines performing complex tasks was Alan Turing. In 1936, Turing proposed a theoretical device called the 'Turing Machine,' which could simulate the logic of any computer algorithm. He later developed the 'Turing Test' in 1950, to determine whether a machine could exhibit human-like intelligence.





The history of AI

The Dartmouth Conference (1956): This is often referred to as the birth of AI as a field. John McCarthy and his colleagues organized the Dartmouth Conference, where they proposed that 'every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it.' The optimism here set the tone for the next few decades of AI research.

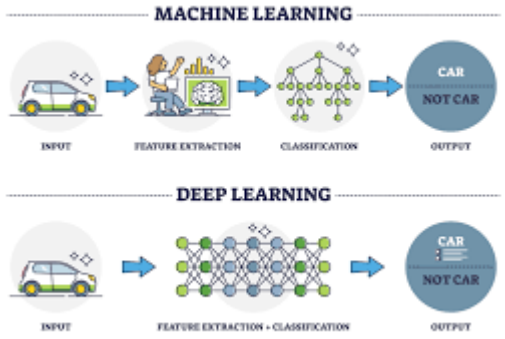
The Logic Theorist: In 1956, Newell and Simon developed 'The Logic Theorist,' considered one of the first AI programs. It proved mathematical theorems and showed that computers could simulate human problem-solving.

- Axiom 1: All humans are mortal.
- Axiom 2: Simon is a human.
- Theorem: Simon is mortal.

The history of AI

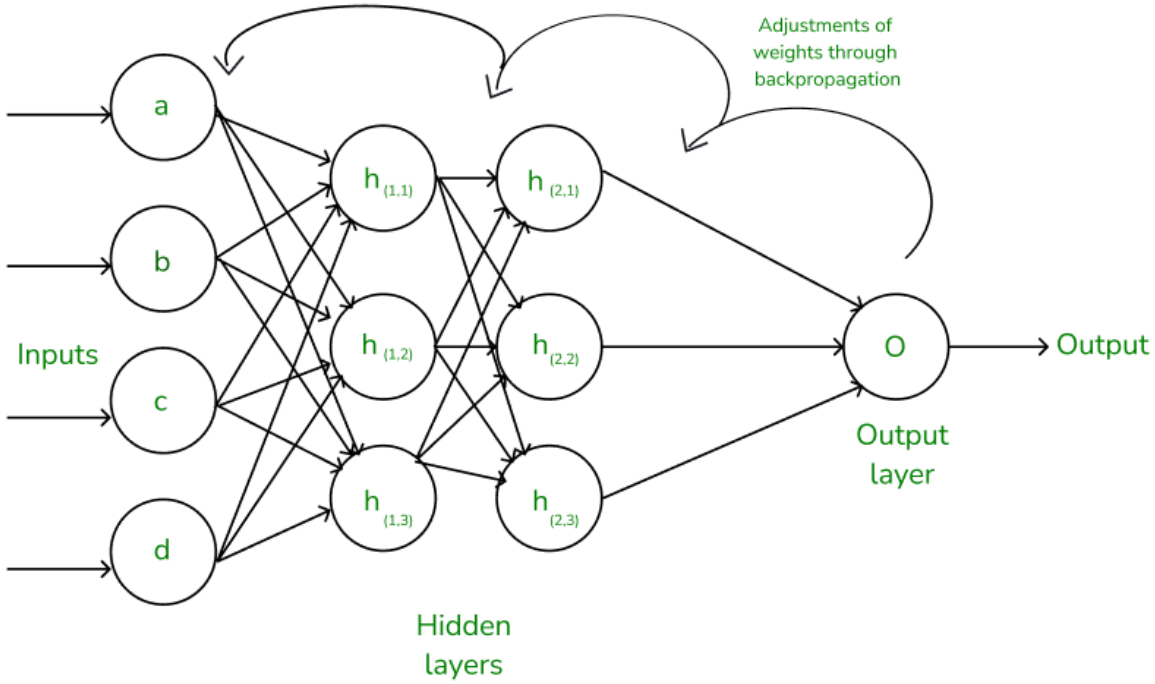
1980s-Neural networks and backpropagation. These models were inspired by the human brain and were more capable of learning from data. A key breakthrough was the 'backpropagation algorithm,' which allowed neural networks to 'learn' from their mistakes by adjusting weights across multiple layers.
Machine learning—machines could now improve their performance over time without being explicitly programmed for every task.

From 2000s-Deep learning, Big Data, GPTs.



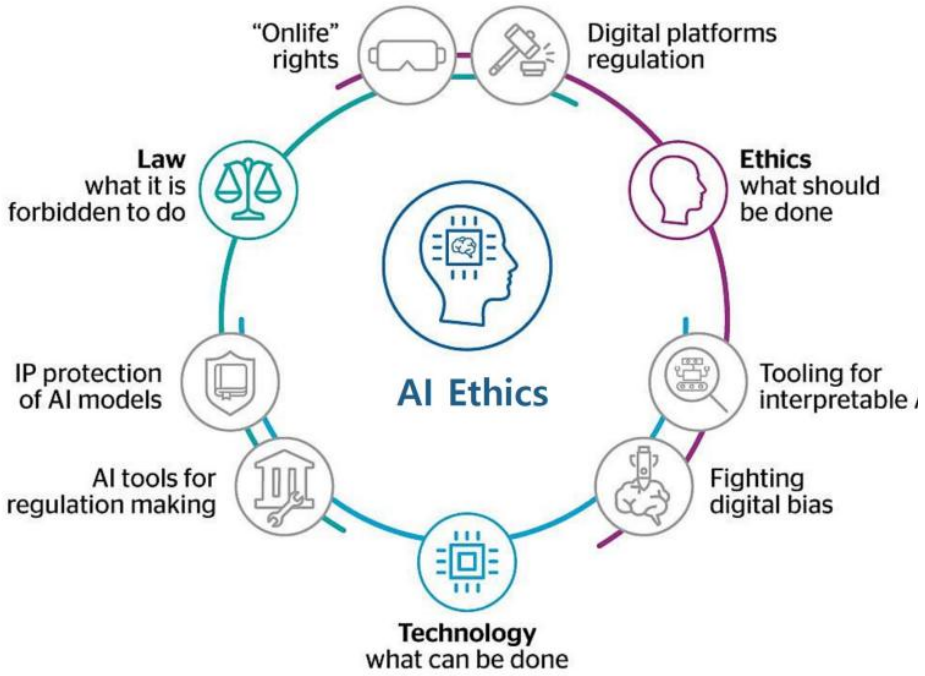
The history of AI

Backpropagation



The history of AI

Ethical Challenges:
Bias in AI models, privacy concerns, and even issues surrounding AI safety.



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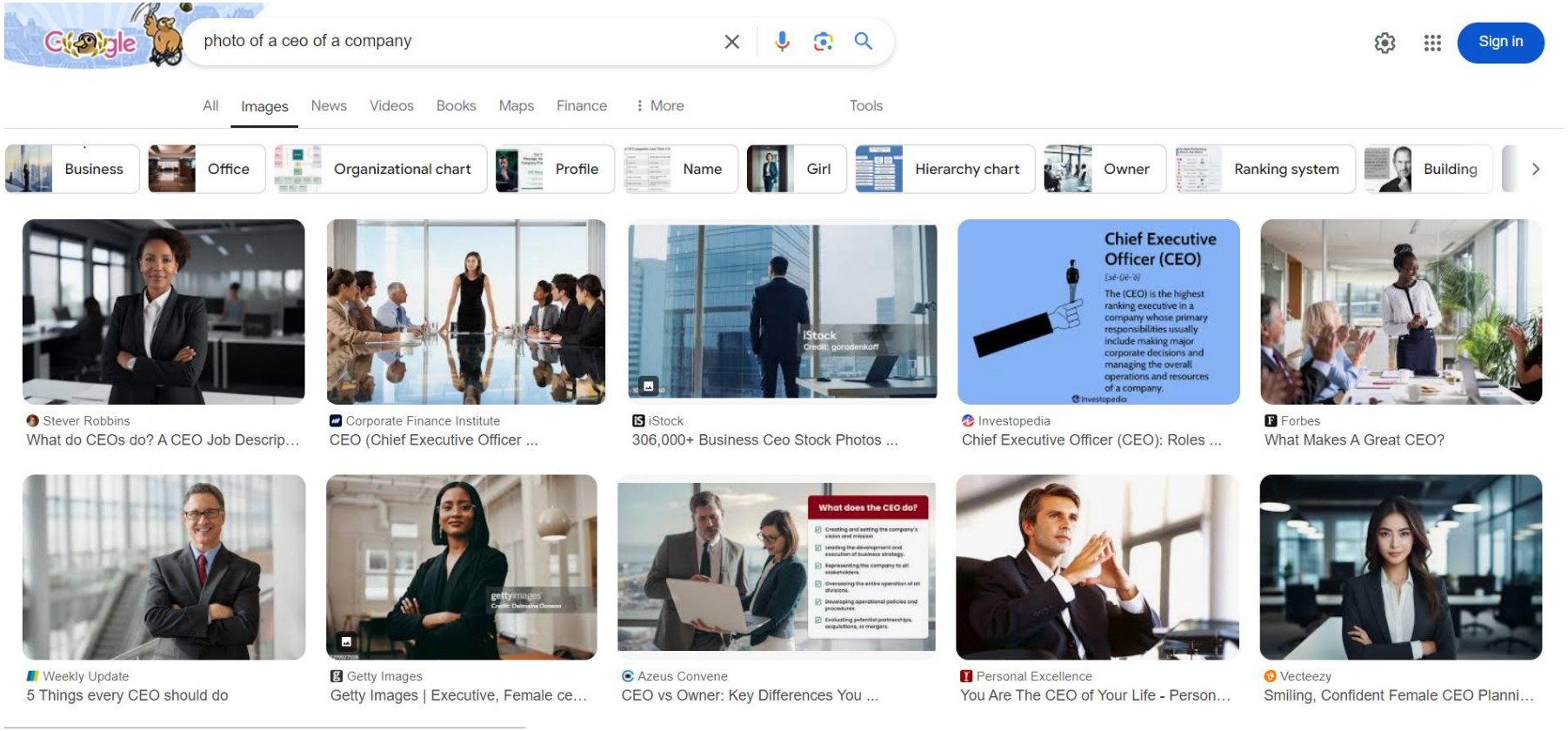
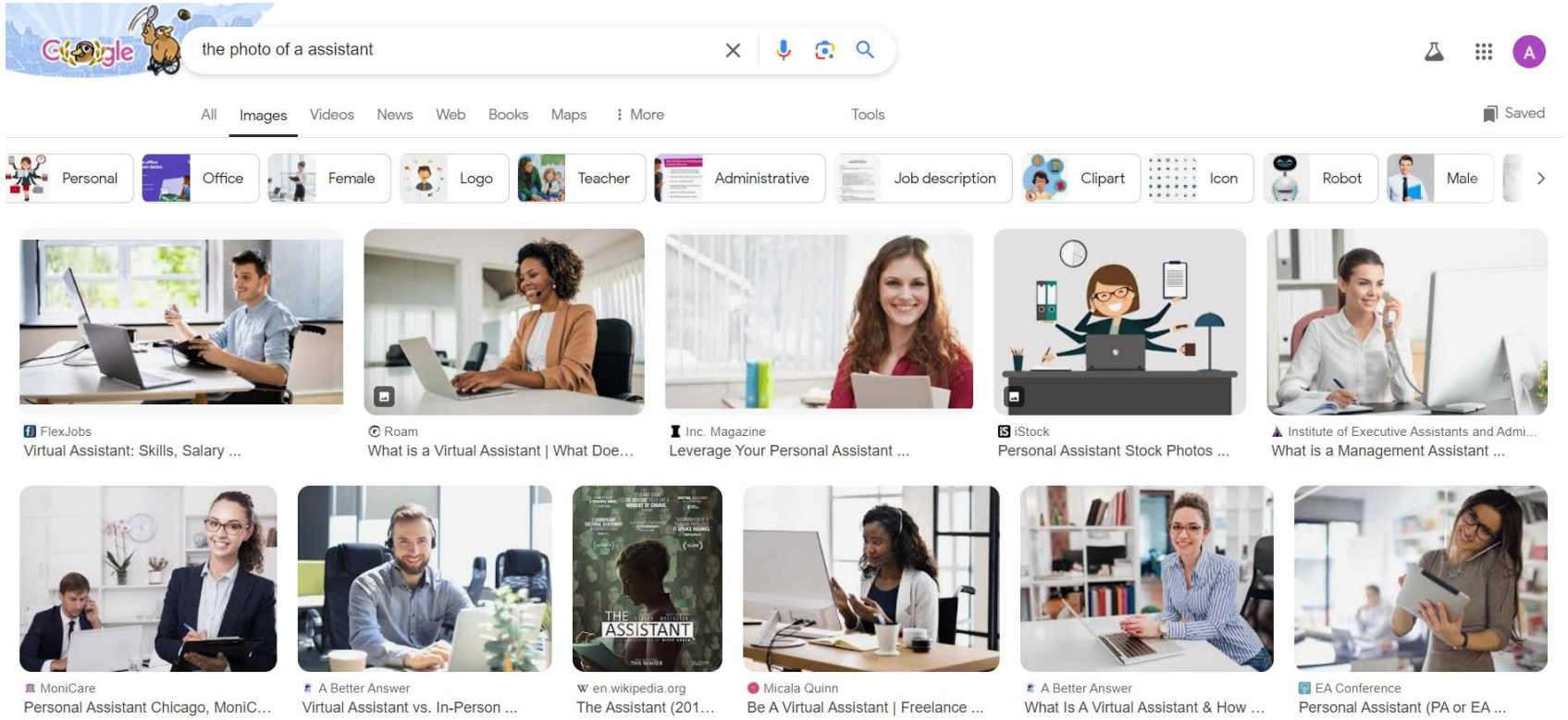


The history of AI



The history of AI

Future of AI:
Researchers are now focusing on areas like explainable AI—making AI systems more transparent and understandable—and general AI, which would have the ability to perform any intellectual task that a human can.